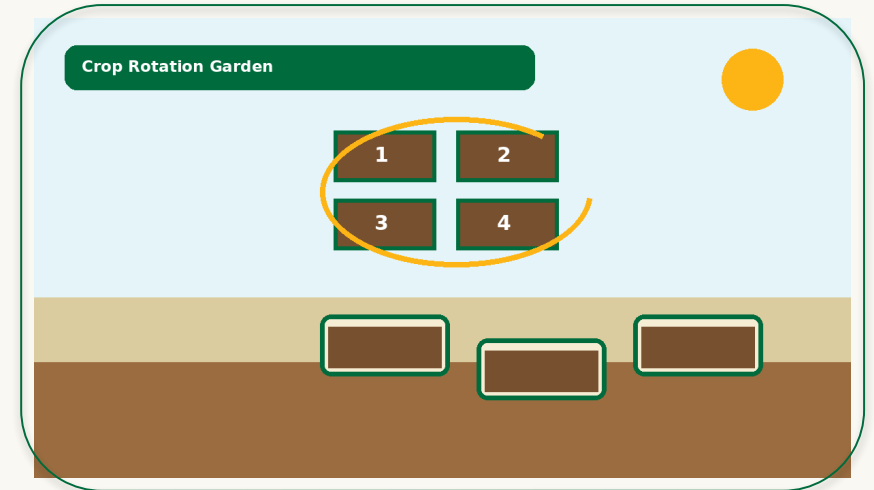


Crop Rotation Garden

Crop Sequence and Soil Management

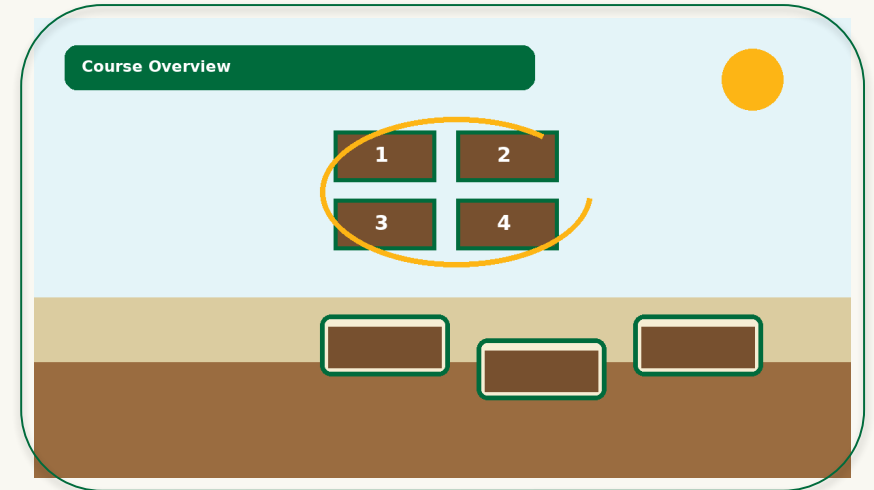
- LIFEWS AgriShine™ GrowMeal™ JSS 1-3 Teacher Training
- Audience: JSS 1-3 teachers, agricultural-science educators and garden facilitators
- Format: Self-paced online course + supervised practical/laboratory orientation
- Recommended duration: 3.5 hours online + 2 hours practical
- JSS learning progression: formulate -> test -> measure -> analyze -> manage



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the production or experimental system; teach all ten technical topics accurately; guide students in planning, measurement and data collection; connect agricultural science with mathematics and evidence; supervise field activities safely; maintain credible records; help students distinguish observation, analysis and conclusion.
- Core materials: teacher guide, plot/system map, crop plan, child-safe field tools, irrigation equipment, labels, clipboards, measuring tapes/rulers, scales and sensors where relevant, data sheets, clean harvest containers, PPE/handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe practical session, and submit a lesson, production plan or experimental protocol appropriate to the course.



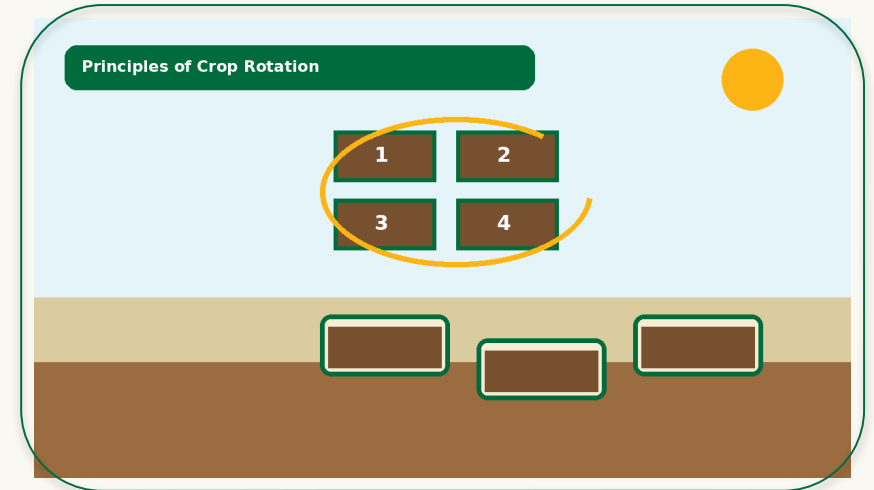
TEACHER PRACTICE

- Training sequence: QUESTION -> PLAN -> IMPLEMENT -> MEASURE -> ANALYZE -> DECIDE.

Principles of Crop Rotation

Teacher knowledge, technical practice and student supervision

- Define rotation as planned sequencing of crops or crop groups through the same land over time.
- Explain goals such as diversifying nutrient demand and disrupting some crop-specific pest/disease cycles.
- Recognize that rotation works with, not instead of, other soil and pest-management practices.
- Use historical bed records to plan future crops.
- Avoid universal rotation recipes without local context.



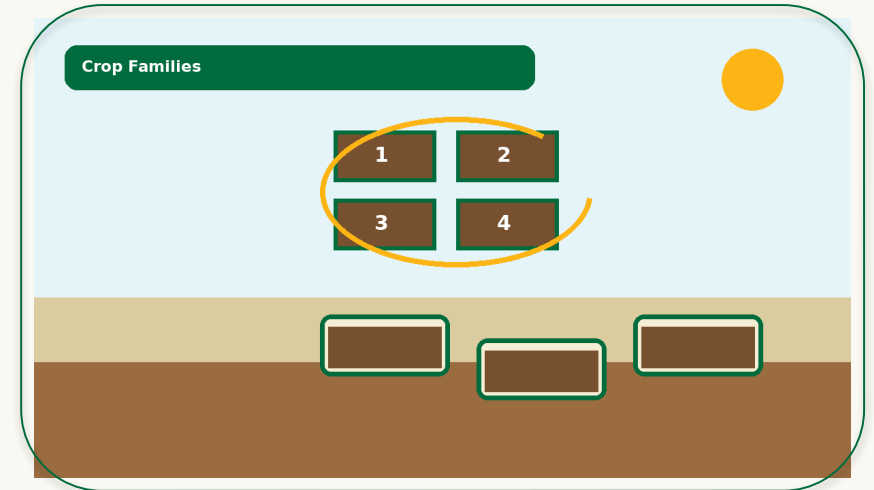
TEACHER PRACTICE

- Review the previous crop history of four beds.

Crop Families

Teacher knowledge, technical practice and student supervision

- Group major school-garden crops into verified botanical families.
- Explain why closely related crops may share pests, diseases or nutrient-use patterns.
- Use family information as one input to rotation design.
- Distinguish botanical family from crop-use categories such as leafy or root vegetables.
- Maintain a crop-family reference chart.



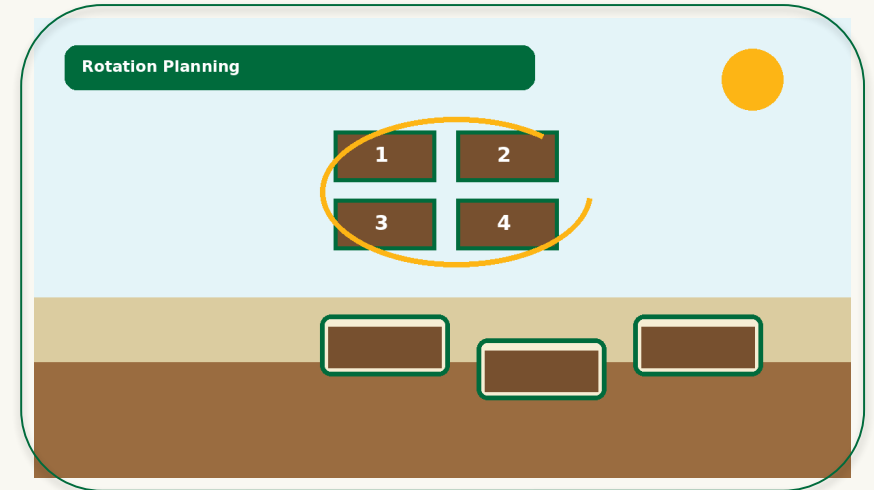
TEACHER PRACTICE

- Prepare a crop-family classification exercise.

Rotation Planning

Teacher knowledge, technical practice and student supervision

- Set the planning period and available blocks.
- List priority crops and their expected durations.
- Sequence crop families while considering season, market/school needs, irrigation and bed availability.
- Include cover crops or fallow periods only where appropriate.
- Revise plans when actual harvest dates differ.



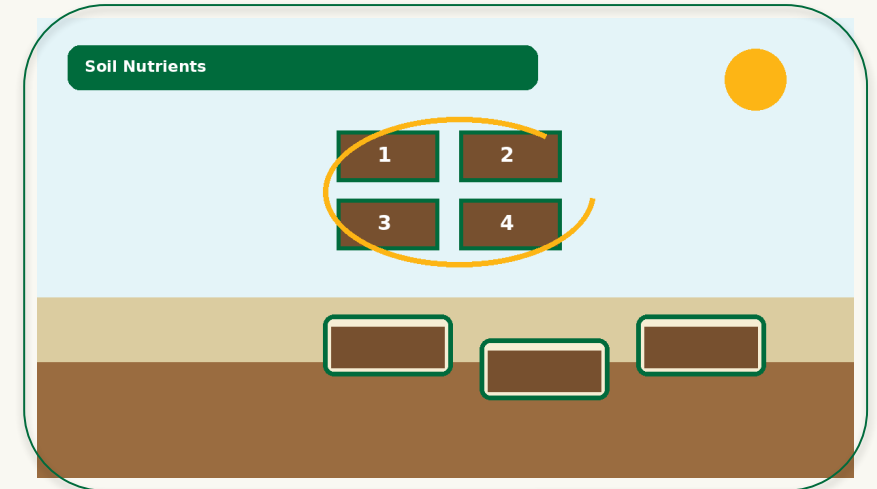
TEACHER PRACTICE

- Design a four-block multi-season sequence.

Soil Nutrients

Teacher knowledge, technical practice and student supervision

- Review nitrogen, phosphorus and potassium as major plant nutrients and recognize other nutrients are also required.
- Explain that different crops can have different nutrient demands.
- Use soil test results where available rather than relying only on visual symptoms.
- Record fertilizer/compost inputs by rotation block.
- Link nutrient management to crop sequence without assuming rotation alone restores fertility.



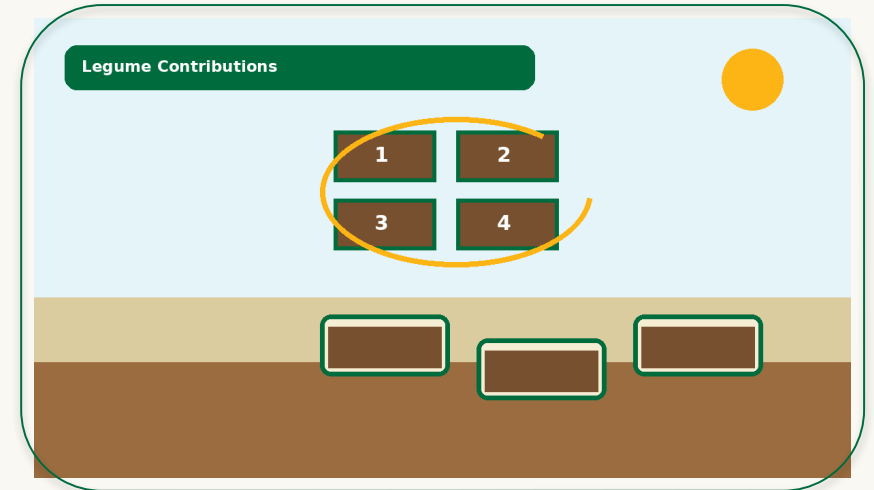
TEACHER PRACTICE

- Add nutrient-management notes to a rotation plan.

Legume Contributions

Teacher knowledge, technical practice and student supervision

- Explain biological nitrogen fixation as a symbiosis between legumes and suitable microorganisms.
- Recognize that fixation varies with crop, nodulation, soil conditions and management.
- Discuss how retained residues or harvested biomass affect what remains in the system.
- Avoid the simplistic statement that any legume automatically fertilizes the next crop.
- Use root-nodule observations only from planned samples.



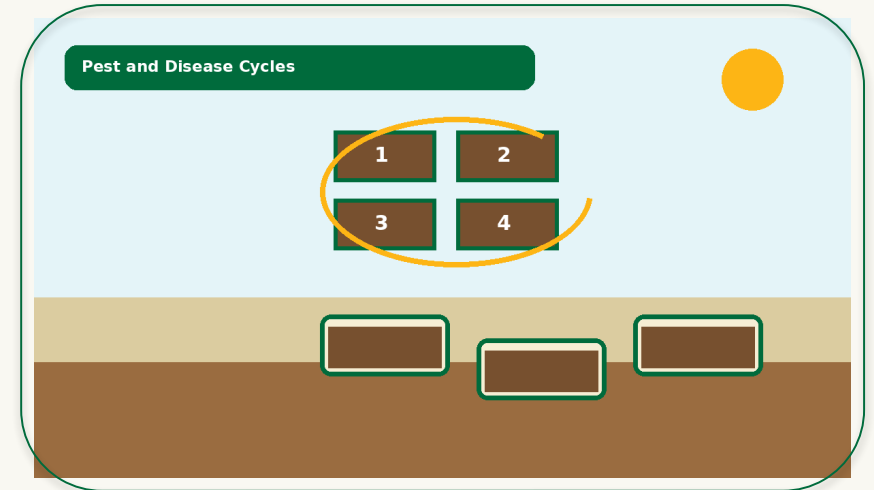
TEACHER PRACTICE

- Prepare a scientifically accurate legume-rotation explanation.

Pest and Disease Cycles

Teacher knowledge, technical practice and student supervision

- Identify examples of crop-specific or family-related pests/diseases relevant to local crops.
- Explain how changing host crops can reduce opportunities for some organisms to persist.
- Recognize pests with broad host ranges may not be controlled by rotation alone.
- Manage volunteer crops and residues appropriately.
- Use integrated pest management alongside rotation.



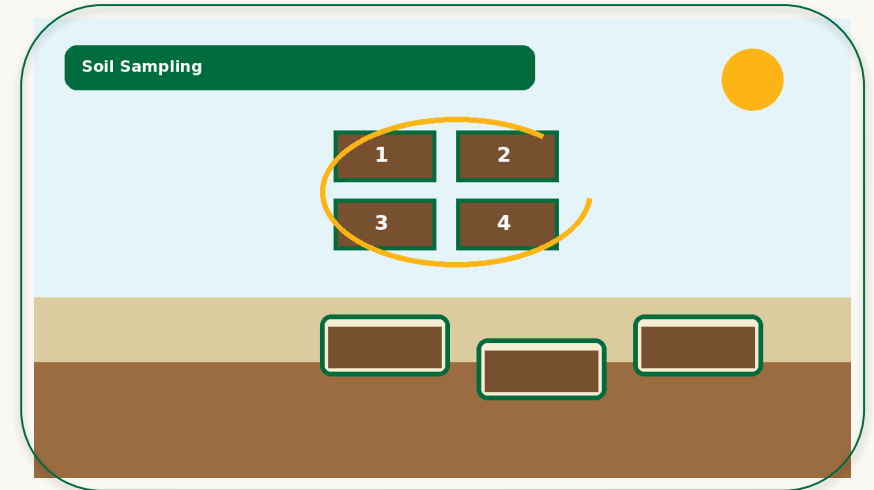
TEACHER PRACTICE

- Map one pest/disease risk across a rotation sequence.

Soil Sampling

Teacher knowledge, technical practice and student supervision

- Define a consistent sampling depth and pattern for the learning objective.
- Use clean tools and avoid contaminated locations unless intentionally studied.
- Combine subsamples only when the protocol calls for a composite sample.
- Label sample by block, depth and date.
- Follow laboratory or field-test instructions exactly if testing is conducted.



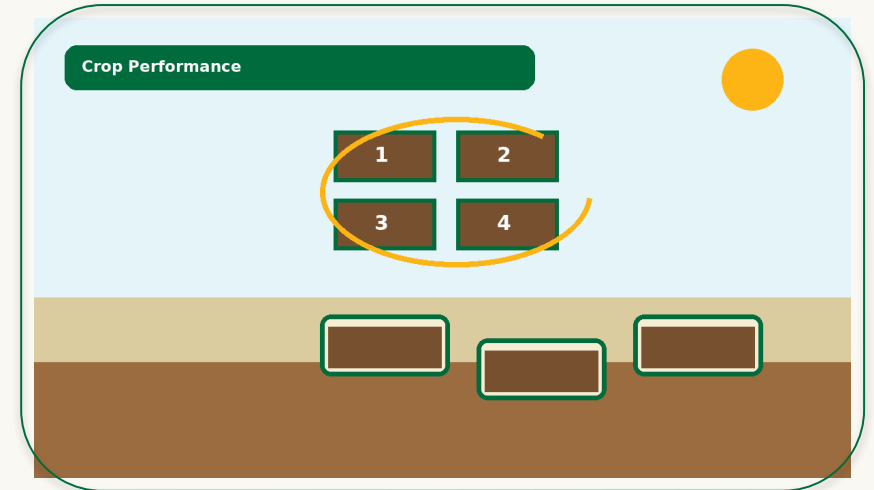
TEACHER PRACTICE

- Demonstrate a simple composite soil-sampling pattern.

Crop Performance

Teacher knowledge, technical practice and student supervision

- Measure emergence/survival, growth, health observations and harvest using consistent methods.
- Compare performance with prior crop history and management records.
- Avoid attributing differences solely to rotation when season or inputs differ.
- Use multiple indicators where possible.
- Record failures and replanting.



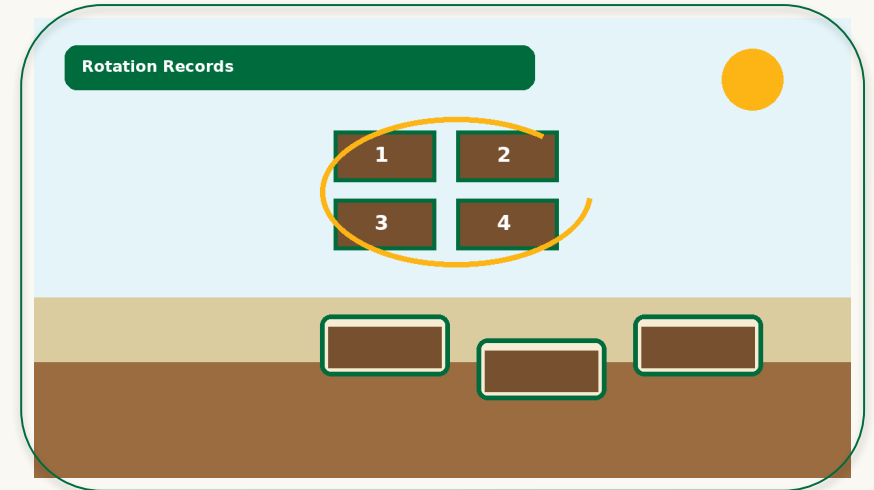
TEACHER PRACTICE

- Create a block-performance scorecard.

Rotation Records

Teacher knowledge, technical practice and student supervision

- Maintain crop family, planting/harvest dates, amendments, pest/disease observations, soil-test information and yield.
- Use stable block identifiers across years.
- Update records after changes.
- Use records before every new planting decision.
- Archive maps and data for future cohorts.



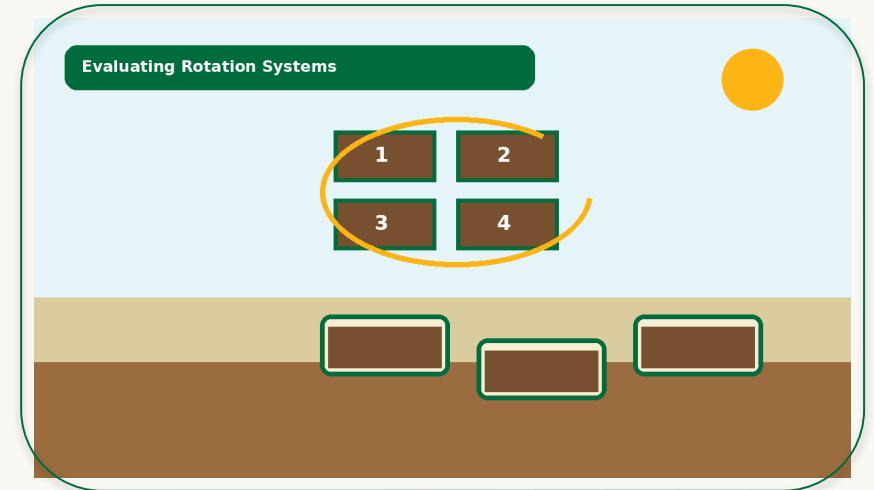
TEACHER PRACTICE

- Prepare a rotation-history database/table.

Evaluating Rotation Systems

Teacher knowledge, technical practice and student supervision

- Evaluate soil indicators, pest/disease observations, crop performance, yield and management practicality.
- Compare more than one rotation cycle where data exist.
- Recognize that strong conclusions require time and consistent records.
- Identify trade-offs between biological benefits and production scheduling.
- Recommend revisions based on evidence.



TEACHER PRACTICE

- Completion task: evaluate and revise a school rotation system.